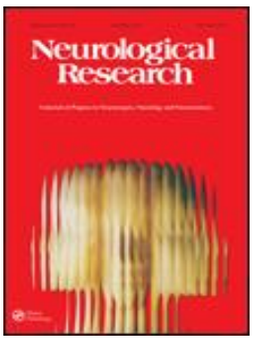

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Neurological Research

A Journal of Progress in Neurosurgery, Neurology and Neurosciences

ISSN: 0161-6412 (Print) 1743-1328 (Online) Journal homepage: http://www.tandfonline.com/loi/yner20

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To cite this article: Albert F. Lawrence & W. Ross Adey (1982) Nonlinear Wave Mechanisms in Interactions between Excitable Tissue and Electromagnetic Fields, Neurological Research, 4:1-2, 115-153, DOI: 10.1080/01616412.1982.11739619

To link to this article: http://dx.doi.org/10.1080/01616412.1982.11739619

Published online: 20 Jul 2016.

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Nonlinear Wave Mechanisms in Interactions between Excitable Tissue and Electromagnetic Fields

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It is now well established that intrinsic electromagnetic fields play a key role in a broad range of tissue functions, including embryonic morphogenesis, wound healing, and information transmission in the nervous system. These same processes may be profoundly influenced by eletromagnetic fields induced by an external force. Tissue exposure to extremely low frequency (ELF) and ELF-modulated microwave fields at levels below those inducing significant thermal effects has revealed highly nonlinear mechanisms as a basis for observed effects. Interactions of phonons and excitons along linear molecules may produce nonlinear molecular vibrations in the form of soliton waves. Solitons exist in a minimal energy state and are extremely long-lived in comparison to linear oscillations. Solitons may convey energy released by chemical reactions from one site to another in enzymes of other long-chain proteins. These nonlinear waves may also couple reaction–diffusion processes in the intracellular and extracellular domains. A model is proposed for interaction between excitable tissue and electromagnetic fields, based on nonlinear waves in the cell membrane, with ionic interactions as an essential step. Calcium fluxes in the extracellular space of the central system are

Neurological Research, Volume 4, Number 1/2
0161-6412/82/010115-00$02.00/0
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modeled by a nonlinear reaction–diffusion system. Membrane molecular solitons may exist in long-chain molecules (Davydov type) and play a significant role in charge transfer; or they may exist as nonlinear waves conveying energy along gel–lipid domains from one protein site to another (Sine-Gordon soliton). Soliton movements occur at subsonic velocities.

KEY WORDS: *Calcium ions, microwaves, nonlinear oscillations, nonthermal, reaction–diffusion process, solitons.*

Introduction

Biological systems ranging from the whole organism to subcellular fragments have been shown to exhibit exquisite sensitivity to low-amplitude electromagnetic fields. Responses to the extremely-low-frequency (ELF) range (0.1–1,000 Hz) and the microwave range (100 MHz–100 GHz) have been of particular interest, partly because of possible implications for the health of industrial and military personnel exposed to these fields in the course of their occupations, and partly because of the challenge to current theories on the interaction between electromagnetic radiation and living tissue. It is necessary to address the latter problem in order to establish which physical and biological processes are relevant to the former. We take up the problem of electromagnetic field (EMF) effects in the particular context of excitable tissue.

Electromagnetic theory is assuming a role in biology far beyond the "classical" applications of chemistry and biochemistry to living systems. It is now well established that essential functions are controlled by weak self-induced electromagnetic fields in a number of tissues of the living organism. The list includes such diverse phenomena as morphogenesis in the embryo, bone remodeling in response to mechanical stress in the mature organism, the healing of wounds in skin and connective tissue, and information transmission in the nervous system. In many cases, the functions of these same tissues may be profoundly influenced by electromagnetic fields induced by external sources.

One of the main difficulties in explaining the biological effects resulting from EMF exposure is the small amount of energy trans-

mitted by a quantum of the incident field, relative to the average thermal energy at the same wavelength present in the biological system. Relaxation time for excitations induced by absorption of quanta of microwave and millimeter-wave radiation are generally considered too short to account for biological effects. In the case of ELF experiments, it is difficult to see how a system could simultaneously show sensitivity to low-level externally induced fields and yet remain stable under intrinsic fields an order of magnitude larger. To get around these problems with thermodynamic models, we must consider mechanisms that involve cooperations coupling or coherence among the constituents of the biological systems. Difficulties with relaxation times lead us to consider energy storage and stabilization mechanisms based on nonlinear interactions among the molecular and submolecular constituents of the system. Observation of amplitude windows (Bawin and Adey, 1976; Bawin et al., 1978; Blackman et al., 1979) as well as frequency windows (Bawin et al., 1975) also indicate nonlinearities in the system under study, at molecular and higher levels of organization.

Present theory concerning nonlinear interactions between radiation and matter allows experimental verification of predictions concerning the mechanisms of the effects of weak electromagnetic fields on living tissue at the molecular level. In our discussion of the physical basis for nonlinear molecular mechanisms we will take as our point of departure the excellent review on nonlinear wave phenomena in tissue-microwave interactions by Taylor (1981). We are especially indebted to him for our introduction to the far-reaching theoretical work of Davydov. In a series of papers concerned with the interaction of phonons and excitons along linear molecules, Davydov and his co-workers have developed a theory of nonlinear molecular vibrations and their significance in biology. These vibrations take the form of soliton waves (a survey of soliton phenomena may be found in the review by Scott et al. [1974]). Due to their existence in a minimal energy state, these solitons are extremely long-lived in comparison to linear oscillations, and could act to convey energy released by chemical reactions from one site to another in an enzyme, an actin molecule, or other long-chain protein. This model has been extended by Davydov to show how charge may be conducted along an alpha-helical protein. One of the most

exciting aspects of the Davydov model for interactions between protein molecules in living tissue and electromagnetic fields is that it predicts resonances in submillimeter wave bands which have been observed in laser Raman spectroscopy experiments (Webb, 1979).

The biological action of an electromagnetic field must be exerted through modifications of cellular chemistry. Although some of the essential steps in the transduction from electromagnetic field to chemical change are likely to be different at different frequencies and for different cellular systems, it seems likely that many transduction processes converge to a common pathway, in that ionic interactions at the cell membrane constitute an essential step in the process. These processes, due to their relatively longer characteristic times maybe responsible for ELF effects observed experimentally.

In subsequent sections of this paper we will show how the classical theory of ionic reaction–diffusion processes and the Davydov soliton model enter into a comprehensive theory of nerve excitation. Our treatment divides into three portions. First, we show that a model based on plasma theory can account for the dynamics of ion movement across the membrane. Second, we show that the inclusion of reaction–diffusion terms in the plasma model can account for threshold phenomena, such as initiation of the action potential. Third, we show how the recent work of Davydov and others leads to a model of the propagation of the action potential along the membrane. All of these processes are coupled to self-generated and externally imposed electrical fields. The model predicts resonances with ambient fields in a number of frequency bands, by establishing a clear connection between events at the quantum level and membrane physiology.

Ionic Effects of Weak Electric Fields

In searching for a link between neural excitability and electric fields, especially at extremely low frequencies, it seems natural to investigate first the relationship between ionic fluxes and cellular processes, since electric fields may directly affect ionic movements. We recall that biological activity of cells in the central nervous

system not only leads to changes in concentrations of neurotransmitter substances, but also to significant changes in extracellular ions. A partial list of the processes involved includes active and passive transport across cell membranes, water movements between various tissue compartments, transport across the blood–brain barrier, and most significant to our argument, the effects of voltage gradients within the tissue (Gardiner-Medwin, 1980).

By reason of their functional role, the most notable of the extracellular ionic fluctuations are changes in K^+ and Ca^{++}. Although a number of effects have been attributed to fluctuations in extracellular potassium, including depolarization of cells, increase of transmitter release, inactivation of sodium conductance, and enhancement of the electrogenic pump (exchange of Na^+ and K^+ across the membrane; Nicholson, 1980), we will limit our discussion to calcium, for the reason that the role of calcium in cell function seems to involve more fundamental processes, many of which are sensitive to small perturbations in local calcium concentrations. Further, the available experimental evidence points to a more direct effect of weak electric fields on Ca^{++} concentration than on other cationic species. Data from recent experiments indicate that the steady-state kinetics of calcium exchange between cell-membrane and extracellular pools are perturbed by the action of electromagnetic fields induced by external sources (Bawin et al., 1975). There is also experimental evidence that electromagnetic fields affect the affinities for calcium of one or more constituents of the cell membrane. Electromagnetic field effects have been shown to depend quite strongly on both the frequency and amplitude of the imposed field (Adey, 1981a, b; Bawin et al. 1975). These effects may take the form of oscillations superimposed upon the usual exponential efflux curves encountered in tracer experiments involving Ca^{++} efflux from cell membrane pools (Bawin et al., 1980). Several types of nonlinear behavior may be observed in tissue systems perturbed by weak electric fields. EMF effects have been shown to arise from fields several orders of magnitude weaker than is needed to cause significant thermal effects in tissue (Adey, 1981a, b). These fields are also weak in comparison to the transmembrane potential shifts necessary to produce the action potential. Given these facts, we expect that effects of the EMFs employed must take place through some

cooperative action of the constituents of the physical and chemical systems involved.

A Plasma Model for Transmembrane Ionic Fluxes

Two recentpapers, Vaccaro and Green (1979) and Triffet and Green (1980), have outlined a theory of the action potential in terms of plasmalike oscillations through the cell membrane. The basis for their work is in a Debye-type model of the cell membrane. In the Debye model, individual ions in the membrane and in the extracellular medium develop an "ionic atmosphere" of oppositely charged ions. For this rason, the Coulomb potential, A/r, of a given ion is modified by a screening factor e^{-br}, so that the expressions for the potential is $\phi = (A/r)e^{-br}$ where r is the distance from the ion. (An excellent account of the Debye theory for electrolytes has been given by Moore [1972]).

The quantity b which is denoted by the term *characteristic length*, is determined by ionic concentrations, temperature, charge, and the dielectric properties of the medium. Under near-equilibrium conditions this quantity may be regarded as the fundamental unit that gives the scale for fluctuations in ionic concentration. The Debye theory predicts that charge layers such as are fixed in the cell membrane will develop counter layers in the adjacent medium.

The model of Vaccaro, Triffet, and Green (which we shall refer to hereafter as the VTG model) proposes a pair of Debye layers at the inner and outer surfaces of the cell membrane of thickness comparable to the characteristic length. In the VTG model, the electric potential and concentrations of ionic species are continuous through the membrane to the adjoining media. In the equilibrium (charged) state, as a result of diffusion processes, a high concentration of potassium ions will accumulate in the cytoplasm adjacent to the inner Debye layer, while a high concentration of sodium ions will accumulate in the corresponding inner Debye layer in the extracellular medium. When an external electric field (such as is produced by a propagating action potential) is applied, there will be plasma oscillations, and the inward movement of sodium ions followed by an efflux of potassium ions will result, thereby sustaining the propagation of the potential. These oscillations are detectable

because the intervention of the cell membrane extends the characteristic relaxation times and Debye lengths several order of magnitude in the membrane. Although the effect is attributed to membrane resistance, the Davydov model of ionic conduction raises the possibility that the effective mass of an ion traversing the cell membrane may be very large. This makes the momentum term in the model equations significant for reasons other than those cited by Triffet and Green.

The effect of the high resistance in the VTG model is not to damp out oscillations, but to extend the effective thickness of the Debye layer across the membrane in the dynamic situation. Within this layer the condition of electrical neutrality does not hold, and concentrations of the charged species may vary according to the local electrostatic field. Since the model implies a dispersion relation between the ions involved (Ca^{++}, Na^{+}, K^{+}, and possibly Cl^{-}) the transient current will take the form of reversible oscillations relative to ions of different species, and a transient current which can be orders of magnitude larger than the ohmic current will develop. In this model the controlling factors are:

1. The interchange of sodium and potassium ions at the inner membrane surface. Due to the chemical potentials involved, this is the major source of energy for the oscillations.
2. The presence of calcium ions. Large variations in chemical potential of calcium ions at the inner membrane surface are possible in the VTG model because resistance to calcium flux is about the same as the resistance to movement of the ions as a whole. This leads to a form of resonance, greatly increasing the amplitude of the overall oscillations.

The Formal Phenomenological Model

The forces which act on a slowly moving ion of species α in or near the membrane can be attributed to the partial pressure, the resistance, and the electric force. We may write these as pressure $= P_\alpha$, resistance $= \theta_\alpha \tilde{u}_\alpha$, and electric force $= e_\alpha \nabla\phi$. Where θ_α is the coefficient of friction for species α, $\tilde{u}_\alpha$is the velocity vector for species α, e_α is the charge for the species α, and Φ is the electrostatic potential.

The principle of conservation of momentum applies, so we may write:

$$m_\alpha \left[\frac{\partial}{\partial t}(n_\alpha \vec{u}_\alpha) + \nabla \cdot (n_\alpha \vec{u}_\alpha \vec{u}_\alpha)\right] \nabla P_\alpha = -n_\alpha (e_\alpha \nabla \phi + m_\alpha \theta_\alpha \vec{u}_\alpha) \quad (1a)$$

where m_α is the mass and n_α is the number density of species α. See Glansdorff and Prigogine (1971) or Davidson (1972)for a derivation of this equation. (This conservation equation differs from that of Vaccaro and Green (1979) in that the hydrodynamic derivative rather than the time derivative is used). The "del dot" term contributes a significant quantity to the equation, since the currents involved are limited to the membrane and its immediate neighborhood, i.e., the gradient is large.

In addition to the conservation of momentum, we must also include an expression for conservation of charge and an expression for the electrostatic potential. These may be written as follows:

$$\frac{\partial n_\alpha}{\partial t} + \nabla \cdot (n_\alpha \vec{u}_\alpha) = 0 \quad (1b)$$

$$\nabla^2 \phi = -\frac{4\pi}{\kappa} \Sigma_d \varepsilon_\alpha \exp(-\beta_\alpha e_\alpha \phi_\alpha) \quad (1c)$$

where Σ_α is the charge density of species α, β_α is $1/kT$ (since the solution is dilute), κ is the dielectric constant, and ϕ_α is the ionic potential of species α. The exponential term in equation (1c) is a result of shielding effects originating in the medium.

If we wish to include the effects of chemical reactions (such as the binding of calcium ions to cell membrane sites) we must add source/sink terms to the conservation of number equations. We write this chemical kinetics term as $R_\alpha (n_1, \ldots, n_m)$ where $(n_1, \ldots, n_m)$ are the number of densities of chemical or chemical species involved in reactions which consume or produce species α. We can assume that membrane-bound ions are immobile, and that reactions that produce or consume free ions do not contribute appreciably to the momentum. If we ignore the momentum, we may write equation (1b) as a system of reaction–diffusion equations, since $\nabla \cdot (n_\alpha \vec{u}_\alpha) = \nabla^2 (D_\alpha \vec{u}_\alpha)$, where D_α is the diffusion coefficient for species

α. Nonlinearities introduced by the chemical kinetics may become important in some cases. We will return to this point in a later section.

Since we are concerned with the effects of induced electromagnetic fields, we must introduce some terms into our model equations to account for their action. Leaving aside the possible effects on chemical kinetics or the conduction properties of the membrane, the action of an electromagnetic field will be felt through a forcing term $F = n_\alpha e_\alpha \vec{\mathrm{E}} + n_\alpha [(\vec{u}_\alpha \times \vec{\mathrm{B}})/c]$ where $\vec{\mathrm{E}}$ and $\vec{\mathrm{B}}$ are the magnetic and electric field vectors of the induced field. We can generally neglect the magnetic vector, and will here, for the sake of simplicity. Given these modifications, our set of model equations becomes:

$$m_\alpha\left[\frac{\partial}{\partial_t}(n_\alpha \vec{u}_\alpha) + \nabla\cdot(n_\alpha \vec{u}_\alpha \vec{u}_\alpha)\right] + \nabla P_\alpha = n_\alpha(e_\alpha \vec{\mathrm{E}} - e_\alpha \nabla\phi - m_\alpha \theta_\alpha \vec{u}_\alpha) \quad (2a)$$

$$\frac{\partial n_\alpha}{\partial t} + \nabla\cdot(n_\alpha \vec{u}_\alpha) = R_\alpha(n_1, \ldots, n_\alpha) \quad (2b)$$

$$\nabla^2 \phi = -\frac{4\pi}{\kappa}\sum_\alpha \varepsilon_\alpha \exp(-\beta_\alpha e_\alpha \phi_\alpha) \quad (2c)$$

Solitons in Plasma Models

In order to gain some insight into the properties of the phenomenological model we have presented in equations (1), we consider the special case of a cold plasma. Although this specialization may be an oversimplification of the physiology, there are some similarities between the VTG model (equations (1) without the "del dot" term in the momentum equations) and a plasma of heavy ions moving in a uniform electron gas. In particular, Vaccaro and Green have shown the existence of nonlinear waves in a special case of the cold plasma equations. If we restrict attention to the region within the membrane in the VTG model, the ionic fluctuations take on dispersive rather than dissipative properties. Similarly, the cold plasma equations may be reduced to a single nonlinear equation with dispersive properties.

Equations (1) may be simplified in the case in which we neglect the pressure and the resistance terms. These conditions correspond to the flow of very heavy ions through a membrane. As we will show, Davydov's model of conduction via nonlinear waves along molecular chains predicts that the effective mass of the moving charge is quite large, so this is a plausible hypothesis.

Since observations of the action potential show that two separate waves of cations cross the membrane, we examine the case of the plasma consisting of two species: electrons and positive ions. In this model, sound wave disturbances are moving with a Mach number slightly greater than unity in a uniform, magnetic-field–free plasma background. (These are not unreasonable assumptions: (1) The sodium and potassium waves move in opposite directions, (2) Experimental observations show the waves may be uncoupled, and (3) The Davydov electrosoliton moves at a velocity greater than the velocity of sound in the medium.) The ions are assumed to be cold and nondrifting relative to the electrons, and electron inertia is ignored. There is considerable overlap between techniques applied by Vaccaro and Green (and also Triffet and Green) and the treatment (following Davidson, 1972) presented here. The basic equations are as follows:

$$\nabla^2\phi = -\frac{4\pi}{\kappa} e\{n_0 [\exp(e\phi/k_B T_e)] - n_i\} \tag{3a}$$

$$\frac{\partial n_i}{\partial t} + n_i \vec{u} = 0 \tag{3b}$$

$$\frac{\partial \vec{u}_i}{\partial t} + \nabla \cdot n_i \vec{u}_i \vec{u}_i = -\frac{e}{m_i} \nabla\phi \tag{3c}$$

where n_0 is the uniform background electron density, and n_i is the ion density. See Davidson (1972) or Bullough (1977). These equations can be specialized to a plane wave of ions propagating along the vector normal to the plane. In our model, this wave originates from the initial disequilibrium condition of the membrane. It begins to propagate when local changes in voltage or membrane phase affect the properties of membrane-bound proteins. We will show

how changes in properties of the membrane-bound proteins are explained by the Davydov model in the final sections of this paper.

The remainder of this section will be devoted to showing how solitary wave phenomena are implied by this model. First we make some simplifications in equations (3). A substitution of variables:

$$(k_B T_e/4\pi n_0 e^2)^{-1/2}\, x \rightarrow x$$

$$(4\pi n_0 e^2/m_i)^{1/2}\, t \rightarrow t$$

$$(k_B Te/e)^{-1}\, \phi \rightarrow \phi$$

$$n_0{}^{-1} n_i \rightarrow n$$

and

$$(k_B Te/m_e)^{-1/2}\, \vec{u}_i \rightarrow \vec{u}$$

yields the dimensionless equations:

$$\nabla^2\phi = \exp\phi - n \tag{4a}$$

$$\frac{\partial n}{\partial t} + \nabla n\vec{u} = 0 \tag{4b}$$

$$\frac{\partial \vec{u}}{\partial t} + \nabla\cdot\vec{u}\vec{u} = -\nabla\phi \tag{4c}$$

Equations (4) specialized to one spatial dimension gave rise to solitary wave solutions. We may either solve equations (4) directly or derive the Korteweg-deVries equation by perturbation techniques. The Korteweg-deVries equation may be written in terms of ion density: $\partial n/\partial t + n\,\partial n/\partial z + \partial^3 n/\partial z^3 = 0$. First we demonstrate a direct solution. We will introduce a moving frame $(x - Mt)$ where $M = V/c$ is the ion sound speed; the boundary conditions are set as: $\phi \rightarrow 0$, $u \rightarrow 0$, $n \rightarrow 1$; $\phi' \rightarrow 0$, $u' \rightarrow 0$, $n' \rightarrow 0$; $x \rightarrow \pm\infty$.

For the argument $(x - Mt)$ we may integrate the system of equations (4) in one dimension. We obtain:

$$\tfrac{1}{2}(\phi')^2 = [\exp\phi + M(M^2 - 2\phi) - (M^2 + 1) + \text{const}] \tag{5}$$

Setting $M-1 = \Delta M$ and assuming that $0 < \Delta M \ll 1$, we may expand (5) and neglect the leading terms to obtain

$$(\phi')^2 = \frac{2}{3}\phi^2 (3\Delta M - \phi) \tag{6}$$

which has the solution

$$\phi = 3\Delta M \operatorname{sech}^2[(\tfrac{1}{2}\Delta M)^{1/2} (x - Mt)] \tag{7}$$

To obtain a Korteweg-deVries equation, we rescale the space and time variables. Since: $(\frac{1}{2}\Delta M)^{1/2} (x-Mt) = (1/\sqrt{2})[\varepsilon^{1/2} (x-t) - \varepsilon^{3/2}t]$ where $\varepsilon = \Delta M$, we define $z = \varepsilon^{1/2}(x-t)$, $\tau = \varepsilon^{3/2}t$. Now expand the quantities in the one-dimensional equations (4a, b, c,):

$$n = 1 + \varepsilon n^{(1)} + \varepsilon^2 n^{(2)} + \cdots$$

$$\phi = \varepsilon\phi^{(1)} + \varepsilon^2\phi^{(2)} + \cdots$$

$$u = \varepsilon u^{(1)} + \varepsilon^2 u^{(2)} + \cdots$$

Further:

$$\frac{\partial}{\partial x} = \varepsilon^{1/2}(\partial/\partial z)$$

$$\frac{\partial}{\partial t} = \varepsilon^{3/2}(\partial/\partial\tau) - \varepsilon^{1/2}(\partial/\partial z)$$

To the lowest order: $\phi^{(1)} = n^{(1)}$
$n_z^{(1)} = u_z^{(1)}$
$u_z^{(1)} = \phi_z^{(1)}$

with solution:

$$\phi^{(1)} = n^{(1)} = u^{(1)} \tag{8}$$

To the next order:

$$\phi_{zz}^{(1)} = \phi^{(2)} + \tfrac{1}{2}(\phi^{(1)})^2 - n^{(2)} \tag{9a}$$

$$-n_z^{(2)} + n_\tau^{(1)} + n^{(1)}u^{(1)} + u_z^{(1)} = 0 \tag{9b}$$

$$-u_z^{(2)} + u_\tau^{(1)} + u^{(1)}u^{(1)} = \phi_z^{(2)} \tag{9c}$$

By algebraic elimination, and use of the first-order solution we obtain:

$$n_\tau^{(1)} + n^{(1)}n_z^{(1)} + \tfrac{1}{2}n_{zzz}^{(1)}$$

This is the Korteweg-deVries equation in $n^{(1)}$. Physically, the term: $n^{(1)}n_z^{(1)}{}_z$ arises from convective effects, and would tend to steepen an initial traveling disturbance until braking occurs. This effect is countered by the dispersive term $\frac{1}{2}n_{zzz}^{(1)}$ which tends to limit the buildup of short-wavelength components. Thus the equation admits nonlinear traveling waves which preserve their identity.

Although a transmembrane solitary wave is not inconsistent with observations of the action potential, it should be noted that the frictional terms leading to dissipative behavior must be included in the model to yield a faithful correspondence with experimental data. The action potential may be regarded as two critically damped Korteweg-deVries (KdV) waves moving in opposite directions across the membrane. The theoretical development in the paper of Triffet and Green is parallel to that of the ion-acoustic wave. In particular, the potential due to potassium and chloride ions is governed by an equation similar to equation (6) in the model of Triffet and Green. This differential equation is pivotal in the derivation of the dynamics of ion flux in both the KdV model and the VTG model, and, under the differing assumptions of the two models, leads to remarkably similar phenomena.

Reaction–Diffusion Systems

We have shown the plasma equations may be used to predict fluxes of ions transverse to the membrane. The second aspect of our model to be considered is the specialization of equations (2) to a system of reaction–diffusion equations. As we have noted, binding of calcium at the membrane surface has significant effects on the properties of the membrane. In addition, since the potential attrib-

utable to fluctuations in Ca^{++} is seen to have decisive effects on the time course of voltage fluctuations within the membrane, perturbations due to reactions between calcium and anionic sites should be greatly amplified in the ionic fluxes.

In general, the equations describing reaction–diffusion systems take the following form:

$$\frac{\partial n_\alpha}{\partial t} = \nabla^2 (D_\alpha n_\alpha) + R_\alpha (n_1, \ldots, n_m) \tag{10}$$

where n_α is the concentration of species α, D_α is the diffusion constant for species α, and R_α describes source and sink terms for reactions that produce or consume species α (Nicholis and Prigogine, 1977). The associated ordinary differential equations:

$$\frac{\mathrm{d}n_\alpha}{\mathrm{d}t} = R_\alpha(n_1, \ldots, n_m) \tag{11}$$

are the kinetic equations associated with equations (10). We must examine the kinetics of calcium at a cell membrane. To show how the reaction–diffusion model may be used to describe initiation of threshold events at the membrane we must examine the kinetics of calcium.

Influx experiments performed with radioactive tracer calcium when total calcium is at or close to equilibrium indicate the existence of an exchangeable membrane pool and a non-exchangeable membrane pool as well as various cytoplasmic pools (Borle, 1969; Borle and Anderson, 1976). Under equilibrium conditions, exchange with the labile membrane pool is 20 times faster than calcium exchange across the cell membrane. Half-lives are on the order of 1.5 min and 30 min for membrane and cytoplasmic pools, respectively, in HeLa cell cultures. Since this study and the majority of all Ca^{++} kinetics studies were made at equilibrium, data on nonlinearities and/or the existence of more than one labile membrane subpool are not available. Evidence on this question has been obtained from other types of experiments. Both membrane-protein–Ca^{++} and membrane-lipids–Ca^{++} interactions have been studied. Nuclear magnetic resonance studies with a variety of Ca^{++}-binding pro-

teins indicate that the calcium binding reaction is rather complex. With the proteins studied (including lysozyme, phospholipase, phosphatase, parvalbumen, troponin-C vitamin D-dependent Ca^{++}-binding protein, and calmodulin) calcium binding was found to occur in two steps with one or two ions bound at each step (Levine et al., 1979; Fullmer and Wasserman, 1979; Seamon, 1980). The first step was usually slow and resulted in major conformational changes in the binding protein, while the second step was much faster and resulted in a general tightening of the protein structure. Thus one might reasonably expect the kinetics of calcium binding to be highly nonlinear at conditions removed from equilibrium.

This hypothesis gains further credence from reports of calcium displacement studies. The function describing the replacement of Ca^{++} from red blood cell membranes by local anesthetics was found to be either hyperbolic or sigmoid, depending on whether the anesthetic agent was cationic or neutral (Low et al., 1979). Further increasing pH was found to change the displacement curves from hyperbolic to sigmoid. The authors propose a model of calcium displacement in which the anesthetic-displaceable Ca^{++} is bound to two adjacent negatively charged lipid head groups on the outer surface of the membrane. The data indicate that neutral anesthetics act in a cooperative manner, through their insertion onto the membrane between adjacent lipid molecules which carry negatively charged binding sites at their head groups. If this model is correct, the major portion of the labile membrane calcium pool would not be bound to protein molecules. Nevertheless, we expect nonlinear effects on Ca^{++} kinetics associated with departures from equilibrium of Ca^{++} concentrations in either case.

A Specific Kinetic Model

Experiments with synaptosome vesicles prepared from rat cerebral cortex show that modulated 450-MHz fields produce changes in Ca^{++} exchange rates (Lin-Liu and Adey, 1980). Since for all practical purposes, metabolic activity is nonexistent in these preparations, one must seek explanations in the physical chemistry of the membrane. We may start with a simple model of binding reactions in which Ca^{++} associates with two anionic sites.

The elementary reactions may be written as follows:

$$Ca^{++} + 2L^- \underset{k_1^-}{\overset{k_1^+}{\leftrightarrows}} L - Ca - L \tag{12}$$

$$Na^+ + L^- \underset{k_2^-}{\overset{k_2^-}{\leftrightarrows}} LNa \tag{13}$$

Where L^- represents an anionic site (such as a phospholipid head group). In order to simplify the argument, we will neglect effects that may arise from the geometry of the lipid bilayer, with binding sites confined to a curved surface. These reactions imply the following kinetic equations:

$$\frac{dc}{dt} = k_1^- C_1 - k_1^+ CP^2 \tag{14a}$$

$$\frac{dN}{dt} = \tilde{k}_2^+ - k_2^- N \tag{14b}$$

where C is $[Ca^{++}]$
P is $[L^-]$
N is $[NaL]$
C_1 is $[L-Ca-L]$
and $\tilde{k}_2^+$ is $k_2^+[Na]$

Since the amount of sodium is large compared with the other ionic species, its concentration does not vary significantly. We may neglect the sodium concentrations in equations (14).

We may determine the homogeneous steady states of the kinetic system by solving the equations:

$$k_1^- C - k_1^+ CP^2 = 0 \tag{15a}$$

$$\tilde{k}_1^+ P - k_2^- N = 0 \tag{15b}$$

given that

$$P + 2C_1 + N = A_1 \tag{16}$$

$$C + C_1 = A_2 \tag{17}$$

The quantities A_1 and A_2 are proportional to the total lipid and total calcium, respectively. We may easily reduce equations (15) to the following relations:

$$C = \frac{k_1^- A_1}{k_1^- + k_1^+ p^2} \tag{18a}$$

$$C = \frac{-\tilde{k}_2^+ A_1 + k_2^- A_2 + 2k_2^- A_1}{-\tilde{k}_2^+ + 2k_2^-} + \frac{k_2^-}{\tilde{k}_2^+ + 2k_2^-} P \tag{18b}$$

Equations (18) may have one or two solutions, depending upon the values of the rate constants and the initial concentrations. In the latter case, the associated reaction–diffusion system may exhibit asymptotically stable waves (Fife, 1977a, b) as well as interesting bifurcation properties.

Thus, the calcium fluxes in the extracellular space of the CNS may be modeled by a nonlinear reaction–diffusion system. Small perturbations may be amplified by nonlinearities of the system. Troy and Field (1977) in their investigation of the amplification of perturbations in a spatially distributed reaction–diffusion system, have found that extremely large effects may arise from small initial disturbances. As the system evolves, chance fluctuations will be amplified and give rise to a large-scale order. This type of order is known as a dissipative structure. In general, where nonlinearities in the reaction kinetics are present, it may be shown that the existence of limit cycles for the kinetic system (11) implies the existence of waves in the reaction–diffusion system (10) (Kopell, 1977, 1978). In particular, subtle modifications of ionic diffusion patterns may be amplified by the nonlinearities of the reaction kinetics, in such a way that weak fields have effects far stronger than simple considerations of relative energies might indicate (Schmidt and Ortoleva, 1977, 1978; Nazarea, 1979). In the VTG model, calcium is a critical factor in the time course of the ionic fluxes across the membrane. Thus, small perturbations in the amounts of free calcium present,

or changes induced by calcium fluctuations in the physical properties of the membrane, will have quite disproportionate effects.

We conclude our review of the influence of Ca^{++} fluxes through extracellular space with a brief discussion of the biological effects of calcium. First, we observe that calcium has a profound effect on membrane excitability. Extracellular calcium ions are indispensable in the maintenance of membrane excitability (Tasaki, 1974, 1975). It is also well established that flux of Ca^{++} in the presynaptic terminal is a critical step in the release of transmitter at a chemical synapse (Nicholson, 1977, 1980; Nicholson et al., 1977). Stimulation of tissue at the surface of the cerebellum has been observed to induce local changes of 20% or more in extracellular calcium ions. These local decreases are paralleled by increases in K^{+} ions. Although the classical Hodgkin-Huxley model accounts for the changes in potassium, it fails entirely to account for the role of calcium. There are several phenomenological models that address the issue of calcium ion movements. Nicholson (1980) attributes changes in extracellular calcium to transmembrane movements. It is at least as likely that binding of calcium to proteins and lipids at the extracellular surface of the membrane is altered during the course of the action potential.

Tasaki (1975) has proposed a model of membrane excitability based on the idea that macromolecules on the external membrane layer possess two stable conformational states with an intermediate unstable state. The concentration ratio between univalent and divalent cations determines whether the macromolecules will be in one of the stable states or in the unstable state. High concentration of Ca^{++} relative to the univalent cations induces the resting state, while displacement by univalent cations at the binding sites induces the depolarized state.

Although this model has received strong criticism as an explanation for spike generation (Keynes, 1970) it may still constitute a valid argument for Ca^{++} modulation of membrane sensitivity. Both protein–calcium and lipid–calcium interactions have been proposed. Adey (1977) has presented a model for extracellular action of Ca^{++} on cell surface glycoproteins. Tubular protein elements within the cell, constituting the cytoskeleton, have membrane connections that function by calcium-dependent mechanisms (Laza-

rides, 1980; Matsumoto and Sakai, 1979; McClain and Edelman, 1978). Membrane excitability has been shown to be affected by concomitant changes in cytoskeletal organization.

A strong case may also be made for lipid–calcium interactions. Calcium has profound effects on artificial membranes. Calcium is found to raise the critical temperature for transition between the gel and liquid crystal states in many bilayer systems (Jacobson and Papahadjopoulos, 1975). The phase of lipids in the plasma-membrane bilayer is significant to the thermodynamics of cell membranes. Membrane-lipid composition is usually metabolically adjusted to ensure phase transition temperatures near the ambient temperature experienced by the cell (Melchior and Stein, 1979).

We may plausibly argue a modulation of membrane excitability through modification of the lipid structure of the membrane by the action of Ca^{++}. The elegant three-way interaction between Ca^{++}, atmospheric pressure, and action of local anesthetics provides additional evidence that this change in structure involves a local phase change. This is exemplified in the following observations. First, calcium binding is opposed by the effects of local anesthetics. (Low et al., 1979). Second, the effects of local anesthetics are reversed by the effects of high pressure (Kendig, 1981).

Modulation of membrane excitability through lipid–calcium interactions is consistent within the reaction–diffusion model presented here. Reconciling the effect of lipid structural changes with the molecular mechanism of ionic conduction is more difficult. In the final sections of this paper we will propose a Davydov-type soliton model for conduction of ions across the cell membrane. This model proposes change conduction along membrane-bound helical proteins which are shown to respond to their chemical and electrical environment. An alternative model of the effect of lipid phase changes on membrane properties has been proposed by Cotterill (1980).

Molecular Solitons

As we have observed, one of the main difficulties in accounting for biological effects of EMF exposure at low levels is the small amount of energy transmitted by a quantum of the field, relative to the

average thermal energy present in the biological system. We have argued that it is necessary to postulate a nonlinearity in the system. In our discussion of reaction–diffusion mechanisms we have devoted some attention to possible cooperative processes, in which a large amount of energy may be released by an expenditure of energy at a low density.

The problems of stability and thermal relaxation may be pursued on this submolecular level. One of the basic questions of bioenergetics is how energy is transferred in biological systems (Green, 1974). A central issue is that the most plausible mechanism of energy transfer (via excited molecular vibrations along biopolymers) is subject to unacceptably short relaxation times. As an example, the amide-I vibration (the C=0 stretch) in protein molecules was presumed to decay to useless thermal energy in less than 10^{-12} sec. This prediction, made on the basis of a linear dynamical analysis, seems to eliminate this mechanism for energy transfer along enzyme molecules.

A potential resolution of this "crisis in bioenergetics" has been proposed by Davydov. He suggests that the nonlinear character of interatomic forces (specifically the hydrogen bond) can lead to the formation of robust solitary waves which exhibit greatly increased radiative lifetimes. This implies a correspondingly increased tendency for molecular vibrations to be the vehicle for energy transfer over long molecular chains (Davydov, 1979a, b).

As a specific application of his idea, Davydov has developed a theory of amide-I vibrations in the alpha-helix protein. According to a linear analysis, energy transferred from amide-I to amide-I bond should be spread out by the effects of dispersion and rapidly become disorganized and lost as an energy source for chemical transformations. In the nonlinear analysis of Davydov, propagation of amide-I vibrations is coupled to longitudinal sound waves of the alpha-helix, and the coupled excitation propagates as a localized and dynamically stable wave called a soliton. The amide-I vibrations are the source of longitudinal sound waves, which in turn provide a potential well that prevents vibrational dispersion. This is the mechanism that holds the soliton together.

A recent numerical study of Davydov's fundamental equations confirms his analytical results (Hyman et al., 1981). A sharp thresh-

old between linear dispersive behavior and nonlinear soliton formation is clearly seen. This threshold is a function of the bond energy coupling between the hydrogen bond stretched by the sound wave and the amide-I bond carrying the excitation energy. It is nearly equal to the energy change in C=0 bond strength per unit length change in hydrogen bond length between formamide molecules. As the coupling parameter increases, another threshold is reached above which solitons will not form. We believe this observation may have profound biological significance.

Each alpha-helix contains three "spines" whose bond sequence is ---HNC=0---HNC=0---HNC=0---, etc. A soliton excitation of an amide-I bond shows an oscillation between C=0 bonds in different spines. This oscillation is at about 500 GHz or 17 per cm. Also, the soliton interacts with the unit cell periodicity of the alpha-helix at a spatial frequency of 125 per cm. Since simple sums and differences of these two frequencies (17 and 125 per cm) exactly reproduce most of the recently measured laser-Raman spectrum seen in metabolically active living cells, there is now strong evidence that Davydov solitons both participate in vital processes and are well described by our model (Scott, 1981). Such solitons on transmembrane (intrinsic) proteins should also display oscillations in the 10-to-170 GHz range as they bounce back and forth from the ends of the molecule.

Davydov has also extended his model to apply to the problem of electron transfer along protein chains. As in the energy transfer model, the key point is the modification of the linear analysis, which holds that when an electron moves, the equilibrium positions of the nuclei remain unchanged. Taking nuclear movements into account leads to the prediction of an "electrosoliton" with the same stability properties as the acoustic soliton (Davydov, 1980). The interaction between the phonon and extra electron carries the electron into a quasi-superconducting state. Should the electrosoliton play a significant role in charge transfer in a biological system, metabolic processes may be tuned to very specific types of variations in the ambient electric field. The results derived from the numerical model of Hyman et al. (1981) and subsequent work by Scott (1981) should apply to the case of an electrosoliton, since the mathematical development in the two cases is exactly parallel. This

implies that the hydrogen bond coupling between the energy levels of the amide-I bond and the phonon energy will be a significant factor in the formation of the electrosoliton. The coupling parameter is evidently a function of the strength of the dipoles involved (C = 0^- and ^+N − H) so an ambient field which acts to polarize (or depolarize) these dipoles will change the coupling parameter. Given fluctuations in field strengths commensurate with voltages observed across biological membranes, one may expect significant changes in this parameter. Such changes will push the dynamics of a helical molecule back and forth across the threshold for soliton formation. Stated another way, a protein molecule might act as a voltage-dependent current gate.

Development of the Davydov Model

Davydov's model of soliton propagation along a protein "spine" is equivalent to the interaction between an exciton and a phonon moving along a one-dimensional crystal lattice (Davydov, 1971, 1979a). Figure 1 depicts this interaction. This may be described in terms of a Hamiltonian for an acoustic soliton:

$$H_{sol} = H_{ex} + H_{ph} + H_{int} \tag{19}$$

where the terms on the right-hand side are the exciton, phonon, and interaction Hamiltonians. The Hamiltonian for exciton is written in operator notation as follows:

$$H_{ex} = \sum_n \left[\left(\varepsilon_0 - \frac{\hbar^2}{m_{ex} z_0^2} \right) a_n^+ a_n + \frac{\hbar^2}{2 m_{ex} z_0^2} (a_{n+1}^+ a_n + a_{n+1} a_n^+) \right. \tag{20}$$

where $\hbar$ is Planck's constant, n is the position of the peptide group along the chain, ε_0 is the internal energy of the exciton, m_{ex} is the effective mass of the exciton, and z_0 is the equilibrium distance of the peptide group. In the operator notation, a_n is the Fermi annihilation operator for an excitation of the amide-I carboxyl bond and a_n^+ is the Fermi creation operator for an excitation of the amide-I bond n.

The Hamiltonian for the phonons is written:

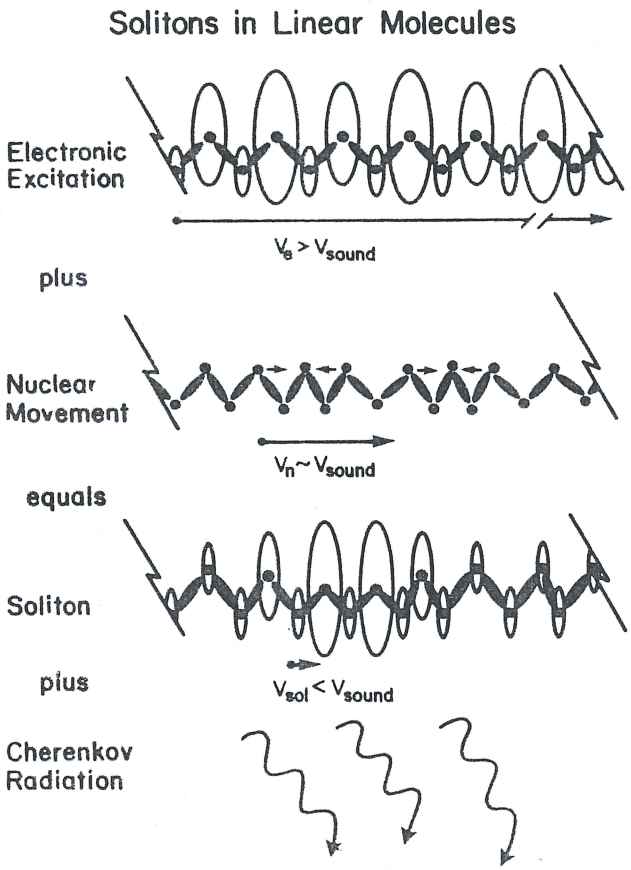

Figure 1. The Davydov model for soliton formation. The "equation" along the left margin illustrates the energy balance in the formation of the soliton wave. The "Cherenkov radiation" in this figure consists of phonons produced by the interaction between the exciton mode and the chain deformation mode. The production of phonons associated with the formation of solitons accounts for the lower energy state of the soliton, and thus its stability.

$$H_{ph} = \tfrac{1}{2}\Sigma_n\left[\frac{1}{M}\hat{P}_n^2 + \chi(u_n - u_{n-1})^2\right] \qquad (21)$$

where $\hat{P}_n$ is the momentum operator of group n, u_n is the displacement of group n from equilibrium, χ is the coefficient of elasticity, and M is the mass of group n.

The interaction Hamiltonian is written:

$$H_{int} = \chi_1 \varepsilon_n a_n^+ a_n (u_{n+1} - u_{n-1}) + \chi_2 \varepsilon_n (a_{n+1}^+ a_n + a_{n+1} a_n^+)(u_{n+1} - u_n) \quad (22)$$

The two sums account for the effect of change of energy of amide-I bonds with respect to deformation and interaction due to resonance as the position of the groups changes. The coupling constants χ_1 and χ_2 are critical in this theory. The properties of the soliton are described in ket notation* by the wave function:

$$\psi_{sol} = \Sigma_n a_n(t)\, c^{\sigma(t)}\, a_n^+ \mid 0 > \quad (23)$$

where $<\psi_{sol} \mid \psi_{sol}> = 1$ is the normalization code, and 0 stands for the ground state. The function $|a_n(t)|^2$ characterizes the distribution of intragroup excitations along the chain. Finally,

$$\sigma(t) = \frac{1}{\hbar} \sum_n [\beta_n(t) \hat{P}_n - \pi_n(t) u_n] \quad (24)$$

Where $\sigma(t)$ is the operator arising from the formal solution of the Schrödinger equation determined by H_{sol}. The functions $\beta_n(t)$ and $\pi_n(t)$ determine the average value of the displacement of molecules from equilibrium positions and the corresponding momenta in the state determined by ψ_{sol}. We have the equations

$$\beta_n(t) = <\psi_{sol}| u_n |\psi_{sol}>, \quad \pi_n(t) = <\psi_{sol}| \hat{P}_n |\psi_{sol}> \quad (25)$$

for the unknown functions $\beta_n(t)$ and $\pi_n(t)$. These may be defined by minimizing the functional $\psi_{sol}| H |\psi_{sol}>$. Since the analytical treatment of the resulting set of coupled nonlinear differential equations is an unlikely prospect, Davydov passes to the continuum limit. This involves letting the number of groups in the chain go to infinity, and the equilibrium distance go to zero. The continuum approximation yields the two equations:

* The "bra" and "ket" notation, invented by P.A.M. Dirac, is standard in quantum mechanics. In this notation, $A|\psi> = \int A\psi d\chi$ by definition. A complete treatment may be found in any recent text. See Marcuse (1980), for example.

$$\left[i\hbar\frac{\partial}{\partial t} - \lambda + \frac{h^2}{2m_{ex}}\frac{\partial^2}{\partial z^2} - 2\chi z_0\frac{\partial\beta}{\partial t}\right] a(z,t) = 0 \tag{26}$$

$$\left[\frac{\partial^2}{\partial t^2} - V_{ac}\frac{\partial^2}{\partial z^2}\right]\beta(z,t) - \frac{2\chi z_0}{M}\frac{\partial}{\partial z}|a|^2 = 0 \tag{27}$$

where the coefficients are appropriately defined from the parameters in the Hamiltonian operators. One may specialize these equations to obtain the solutions corresponding to excitations moving along the chain with velocity less than the velocity of sound. These solutions correspond to solutions of the nonlinear Schrödinger equation for (z, t):

$$\left[i\hbar\frac{\partial}{\partial t} - \Lambda + J\frac{\partial^2}{\partial z^2} + G|a(z,t)|^2\right] a(z,t) = 0 \tag{28}$$

where, as before, the coefficients are defined in terms of the parameters appearing in the Hamiltonians. This equation has soliton solutions:

$$a(z,t) = \frac{\sqrt{z_0 g}\,\exp\{i[k(z-z_0)-wt]\}}{\sqrt{2}\cosh\,[g(z-z_0-vt)]} \tag{29}$$

The effect of this propagating wave is to transfer energy along a protein and helical molecule as in Figure 2. An alternative approach, more suited to numerical modeling, is based on the calculation of a functional for the Hamiltonian operators. This method may be applied to a model which takes account of the three spine configuration of the protein alpha helix. This results in the following system of ordinary differential equations:

$$\begin{aligned} i\hbar\frac{da_{n\alpha}}{dt} &= [E_0 + W + \chi_1(\beta_{n+1,\alpha} - \beta_{n-1,\alpha})]a_{n\alpha} \\ &\quad - J(a_{n+1,\alpha} + a_{n-1,\alpha}) + L(a_{n,\alpha+1} + a_{n,\alpha-1}) \\ &\quad + \chi_2[(\beta_{n+1,\alpha} - \beta_{n\alpha})a_{n+1,\alpha} + (\beta_{n\alpha} - \beta_{n-1,\alpha})a_{n-1,\alpha}] \end{aligned} \tag{30a}$$

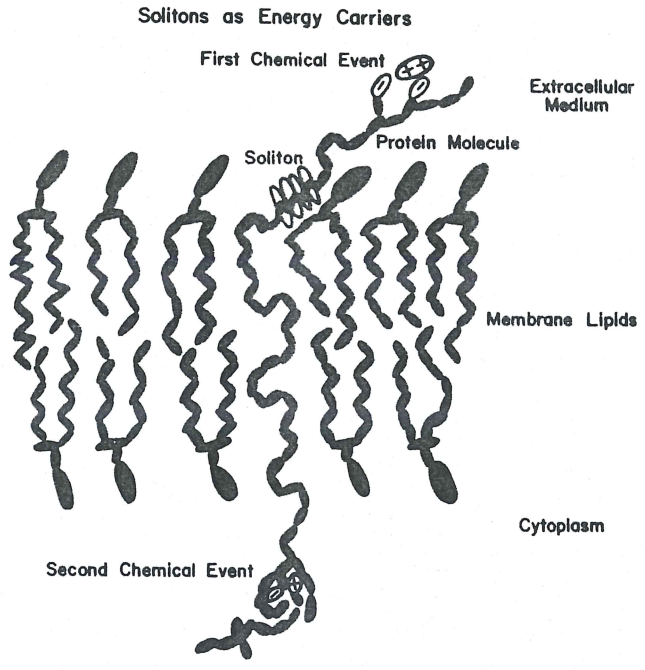

Figure 2. The biological role of solitons. A hypothetical chemical event produces energy at one end of a long-chain molecule. The energy is transferred in the form of a soliton along the molecule and induces a secondary chemical event at the other end.

$$M\frac{d^2\beta_{n\alpha}}{dt^2} - w(\beta_{n+1,\alpha} - 2\beta_{n\alpha} + \beta_{n-1,\alpha}) =$$

$$\chi_1(|a_{n+1,\alpha}|^2 - |a_{n-1,\alpha}|^2) \qquad (30b)$$

$$+ \chi_2[a^*_{n\alpha}(a_{n+1,\alpha} - a_{n-1,\alpha}) + (a^*_{n+1,\alpha} - a^*_{n-1,\alpha})a_{n\alpha}]$$

$$W \equiv \frac{1}{2}\sum_{n,\alpha}\left[M\left(\frac{d\beta_{n\alpha}}{dt}\right)^2 + w(\beta_{n\alpha} - \beta_{n-1,\alpha})^2\right] \qquad (30c)$$

where equation (30a) describes the propagation of amide-I vibrations via dipole–dipole interactions and (30b) describes the propagation of longitudinal sound waves. The total longitudinal sound energy is defined by equation (30c) (Hyman et al., 1981). In this model n corresponds to the turn number along the alpha-helix and

to the spine number. This model has been used as the basis of computer models of soliton propagation along the alpha-helix (Hyman et al., 1981). The only variable parameters are χ_1 and χ_2, the rest being determined from basic principles. Since terms involving χ_2 do not have a strong effect, the computer models show that the value of χ_1, on the other hand, determines whether solitons will form.

We may extend the model described by equations (20), (21), and (22), to account for the interaction of soliton excitation with an ambient electromagnetic field. We may describe this interaction in terms of the Hamiltonian:

$$H'_{\text{sol}} = H_{\text{sol}} + H_{\text{sol-field}} + H_{\text{field}} \tag{31}$$

where $H_{\text{sol-field}} = H_{\text{ph-field}} + H_{\text{ex-field}}$. The terms $H_{\text{ex-field}}$ and $H_{\text{ph-field}}$ describe exciton–field and phonon–field interactions. We can assume that the field is not changed appreciably by interaction with the protein molecule, so we may write:

$$H'_{\text{sol}} = H_{\text{sol}} + H_{\text{sol-field}} \tag{32}$$

This simplified Hamiltonian may be treated by perturbation techniques. A full treatment of the effects of an ambient electrical field will be presented in a subsequent publication.

The model for charge transfer is similar to that for transfer of amide-I energy. We return to the one-spine model Davydov electrosoliton (Davydov, 1980). In this model we write $H_{\text{es}} = H_{\text{el}} + H_{\text{ph}} + H_{\text{int}}$, where H_{el}, H_{ph}, and H_{int} are the Hamiltonians for an electron, a phonon, and the electron–phonon interaction. The term for the phonon may be defined as before, but a more precise approximation includes anharmonic terms in the phonon Hamiltonian. The Hamiltonian for the electron is described as follows:

$$H_{\text{el}} = \sum_n [(\varepsilon_0 - \mathrm{D}_0)\, b_n^+ b_n - \mathrm{D}_\alpha (b_{n+1}^+ b_n + b_n^+ b_{n+1})]$$

where ε_0 is the energy of an electron in an isolated group, and b_n^+ is the operator describing the presence of an electron at the group at

position n. The operators D_0 and D_α are defined as follows: Let $\psi = b_n^+|0>$ be the representation of the wave function of an electron in an isolated amino acid in the ket notation. If $W(\vec{r}-n\vec{x}_0)$ is the operator of the potential energy of an electron having coordinate $n\vec{x}_0$ interacting with a neighboring site $(n+1)\vec{x}_0$, then

$$D_0 = \int |\psi(\vec{r}-u\vec{x}_0)|^2[W(\vec{r}-n\vec{x}_0+\vec{x}_0) + W(\vec{r}-n\vec{x}_0-\vec{x}_0)]\, d^3r$$

$$D_\alpha = \int \psi^*(\vec{r}-n\vec{x}_0-\vec{x}_0)W(\vec{r}-n\vec{x}_0-\vec{x}_0)\psi(\vec{r}-n\vec{x}_0)d^3r$$

These two operators characterize the energy of interaction of an electron near site $n\bar{x}$ with neighboring groups of the chain, and the energy of the exchange transport of the electron between neighboring groups. In the case of an electrosoliton with only harmonic terms, the derivation of electrosoliton equations proceeds parallel to the derivation for the acoustic soliton equation (28). The more interesting case includes cubic anharmonicities. Davydov has proven that in this (more general) case the propagation velocity will exceeed the speed of sound. This justifies one of the assumptions we have made in our treatment of the ionic flux through the membrane. The effective mass of the electron is shown to be increased by an amount proportional to the molecular weight of an amide group. This justifies the assumption made on the effective mass of the ion in our plasma model.

There are reasons to believe in effects other than direct soliton–field interactions. One candidate for such an effect is the creation of a standing compression wave across the cell membrane. A compression wave would arise in the 50-GHz region because of the difference in potential across the membrane. This wave will couple with phonons in intrinsic membrane proteins. The main effect of such an interaction would be to alter the threshold for formation of solitons along the proteins. Another candidate for an indirect interaction is through the polar head groups of membrane lipids. Nonlinear waves may be induced in the membrane and these would interact with the membrane-bound proteins through Coulombic forces. Such interactions would also alter soliton propagation along proteins.

The Hamiltonian for molecules in a lipid membrane is quite com-

plex, even in comparison to the Hamiltonians that have been written for the protein alpha-helix. In the most general situation, terms must be written for both the hydrocarbon tail groups and the polar head groups. As a first approximation we will use a model that assumes that the hydrocarbon tails are immobile. Since lipid membranes of living cells are believed to contain patches of lipids in the gel state, in which the tail groups are immobile, this is a reasonable model to use (Melchior and Stein, 1979).

Lipids in a membrane in the gel state form a crystalline two-dimensional lattice, the head-group dipoles in the plane of the membrane. We may thus write the Hamiltonian for a single dipole head group in terms of rotational variables for the dipole, and six dipole–dipole interactions with the lipid's nearest neighbors. This two-dimensional generalization of the classical Toda lattice is solvable by the inverse scattering method, so we expect that our model will exhibit interesting regularities of behavior. Dipoles in the membrane will interact with lateral compression waves (phonons). The Hamiltonian for dipole–dipole, phonon, and phonon–dipole interaction is the two-dimensional analogue of Davydov's Hamiltonian for acoustic solitons.

Conclusions

The model that has been the standard of conventional wisdom in the field for 30 years, that of Hodgkin an Huxley (1952), suggests that certain membrane molecules (most likely proteins) act as voltage-dependent gates for specific ions and that these ion-specific molecules exhibit a kinetic dependence on the chemical potentials of their respective ionic species. Unfortunately, intensive biochemical research during this same period has failed to isolate specific membrane molecules with the third- or fourth-order kinetics required by the model. The Hodgkin-Huxley model suffers from other inadequacies, also related to the mechanism of ionic conduction. No provision can be made in the strict context of the model for the role of calcium in the maintenance of membrane excitability, nor is there any indication of how lipid-soluble anesthetics, which do not interact strongly with membrane proteins, can affect ionic movements.

By introducing nonlinear molecule dynamics into the classical phenomenological account of ionic reaction–diffusion systems, we have derived a comprehensive model which can account for initiation and propagation of the action potential. This model is presented in terms of basic physical principles and admits inclusion of EMF effects at several levels. These include field effects of transmembrane fluxes, on the propagation of extracellular disturbances, and on the dynamical properties of membrane-bound molecules.

Our model of membrane excitability rests upon three basic principles. First, the membrane proteins serve to conduct the ions across the membrane. Second, the interactions of lipids and calcium in the production of a particular thermodynamic phase in the membrane is essential to the propagation of the excitation. Third, nonlinearities in the molecular dynamics of the system, rather than higher-order chemical kinetics, are the key to the process (Figure 3.)

We have based our model partly on extensions of existing theory for which we have no direct evidence. At this point we pose several theoretical problems. The most important is that Davydov's model of conduction of electrons and energy along alpha-helical proteins may be extended to cover the case of conduction of positive ions along helical molecules in the membrane. This assumption is consistent with the ion-channel model proposed by Urry (1978).

A problem arising from this model is the extent to which solitons can be scattered by an electromagnetic field from inhomogeneities in protein molecules. (This hypothetical event is depicted in Figure 3). A question of particular interest is whether energy from the absorption of several photons at submillimeter wavelengths might be stored in a soliton. The persistence of the soliton vibration makes this a plausible speculation, consistent with the numerical studies of Scott et al. (1981). Multiple photon events in the retina have been reported in studies with infrared radiation at 800 to 1355 nm (Dmitriev et al., 1979). The biological consequences of a hypothetical nonlinear scattering process producing a soliton are illustrated in Figure 4. Numerical models of the type tested by Hyman et al. (1981) have predicted far infrared resonances actually observed in the laser-Raman spectroscopy of living organisms. Davydov's work requires extension to yield a full quantum mechanical treatment of EMF interactions with alpha-helix proteins.

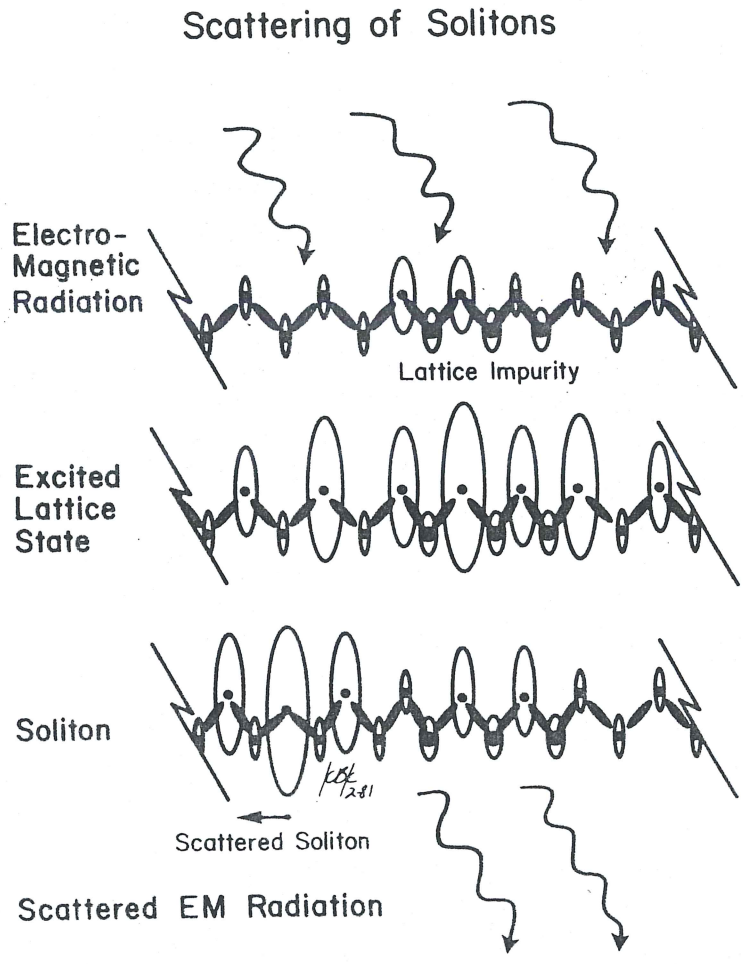

Figure 3. Multiphoton processes in inhomogeneous linear molecules. In a hypothetical interaction, photons in the UHF to infrared range may be absorbed by a long-chain molecule. Formation of solitons may be induced by "lattice impurities," such as peptide groups of anomalous composition in a protein chain. Excess energy may be radiated as photons as well as phonons.

The final concept to be considered here is whether nonlinear waves can convey energy along gel–lipid domains in the membrane from one protein site to another. The actual physical situation in a sheet of dipole head groups in a biological membrane is more complex, in that rotations have several degrees of freedom. We note,

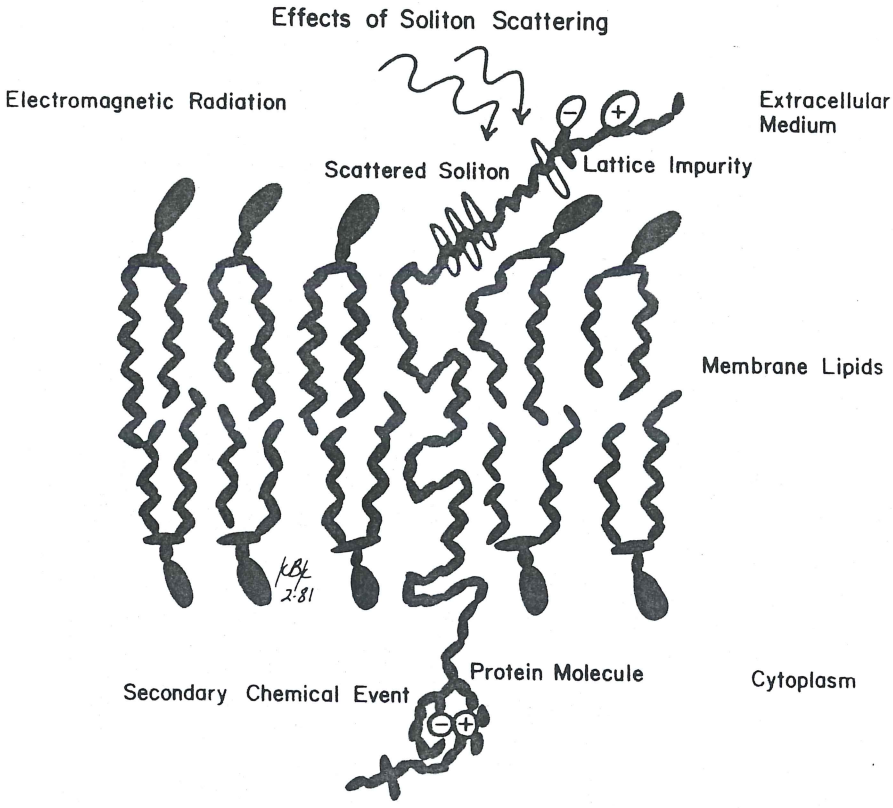

Figure 4. Biological consequences of inelastic scattering processes. In this hypothetical event, one or more photons are absorbed in a long-chain molecule inducing a soliton wave which plays the same role as a soliton produced by a chemical event.

however, that the chemical bonds constrain the mode of rotation of the polar head groups in an axis parallel to the plane of the membrane (Figure 5). Further, the packing of lipids in gel-type domains should impose a preferred orientation on the dipole head groups, giving the system a one-dimensional character. The dynamics of the resulting nonlinear system may be approximated by the equation $\phi_{xx} - \phi_{tt} = \sin\phi$, where ϕ represents the rotation of the head group away from the preferred orientation. This system exhibits solitary waves. The function of such solitary waves would be to transfer energy along the surface of a lipid membrane, as is shown in Figure

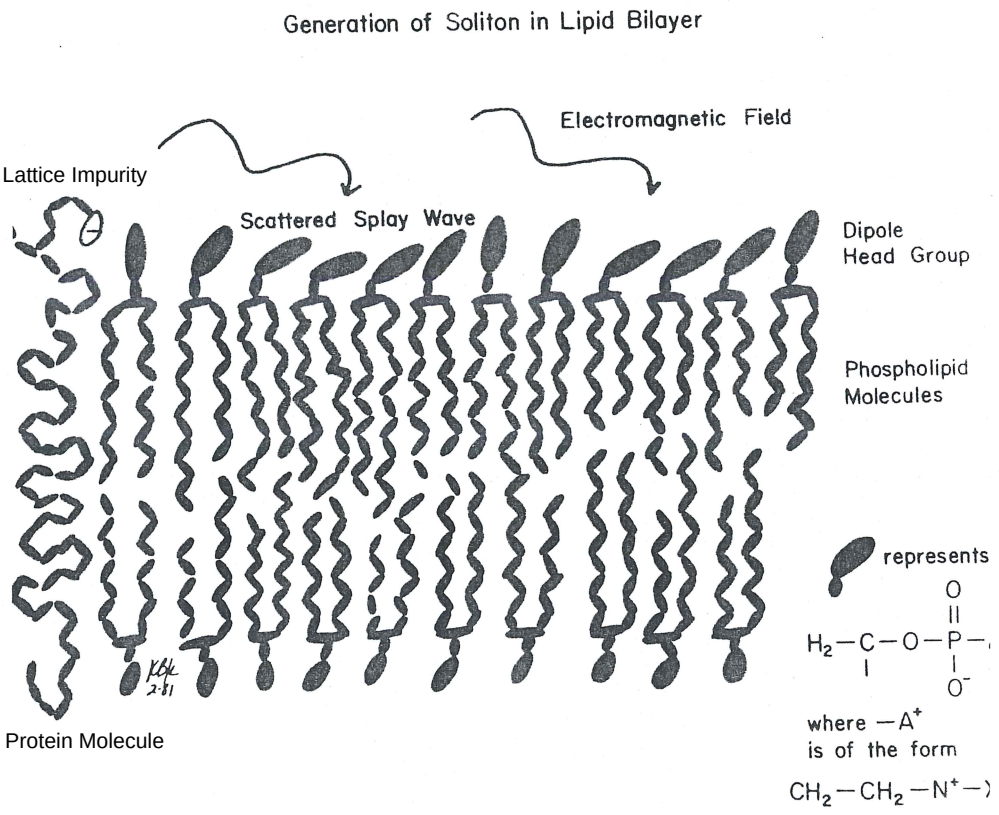

Figure 5. Splay waves in the orientation of head groups of membrane phospholipids. Normally, the dipoles are oriented parallel to the plane of the membrane, but for purposes of the figure, the head groups are rotated into the plane of the section. Splay waves satisfy the Sine-Gordon equation; thus, a lipid layer should exhibit both extended waves and solitons. The wave illustrated is induced by an ambient electromagnetic field.

6. These nonlinear waves may play a significant role in the thermodynamics of a biological membrane and serve as an alternative model to the Davydov soliton.

Some of the main target areas for further research in the biological role of nonlinear waves are shown in Table 1. This table outlines a program for study of bound energy states in biological systems. Trullinger (1978) has elegantly summarized the present situation:

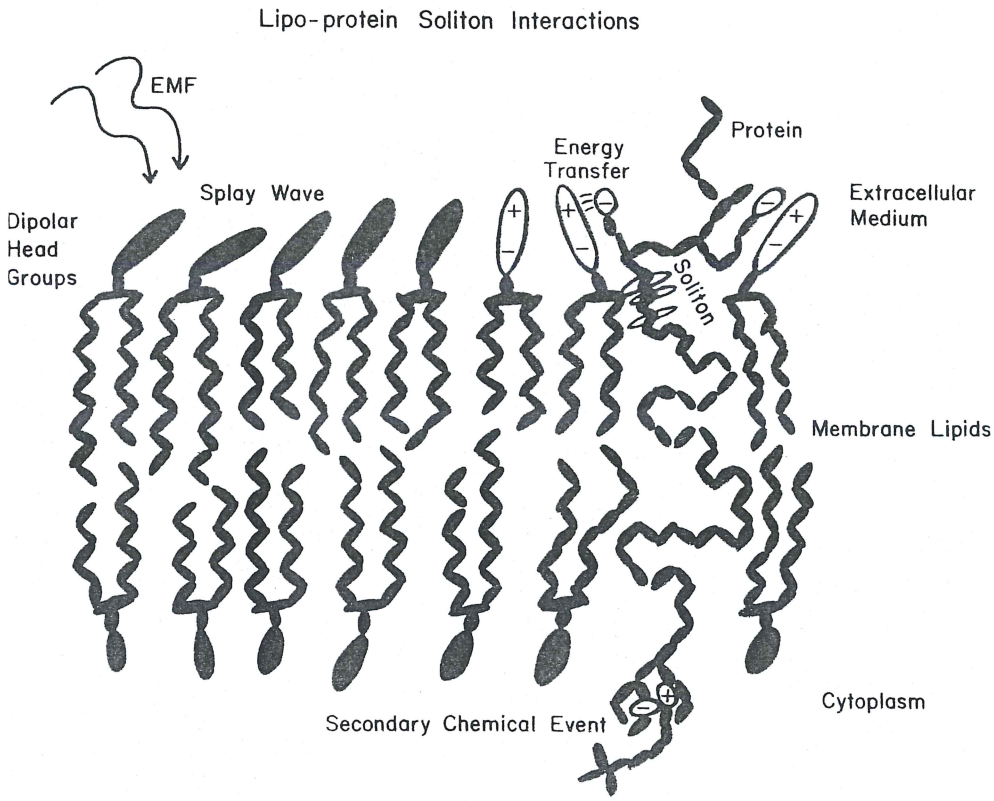

Figure 6. Splay waves as an energy transfer mechanism in a biological membrane. This figure illustrates a hypothetical interaction between and EMF-induced wave in the membrane and a membrane protein. The energy of a splay wave traveling along the membrane is transferred to a protein via Coulombic coupling between dipole-type head groups and polar sites along the protein chain.

Once the uninitiated can bring himself to revolt from his "linear" upbringing (via the harmonic oscillator and Schrödinger's equation), and accept the fact that large-amplitude localized objects can exist which *do not* spread in time, then he stands on the threshold of developing that wonderfully simple and beautiful pattern of thought that emphasized the remarkable stability of such objects, their very natural use as elementary excitations or as fundamental objects present in the ground state, their coexistence in many cases with extended linearized solutions, their striking

particle-like behavior in the presence of perturbations, and perhaps most important of all, their essential role in describing so many physical phenomena that *cannot* be explained in any other way.

Table 1.
Solitons in Biological Systems

Physical System	Electrochemical	Dipole Layer	Molecular
Soliton Phenomena	Ion-acoustic waves	Lattice waves	Subacoustic excitations
Conjectured Sensitivity	Extremely low frequency DC-audio	Microwave-IR light	Millimeter microwave infrared
Model Equation	Kortweg-deVries	Sine-Gordon	Non-linear Schrödinger
Mechanism	Field-coupled reaction-diffusion systems	Rotation of dipoles	Coupled molecular excitations and nuclear motions
Transduction	1. Field-induced extracellular ionic waves 2. Change of threshold of soliton formation	Phonon-photon coupling	Resonances with dipole vibrations in protein alpha-helix
Experimental Test	Observation of change of optical properties via laser interferometry	Laser raman spectroscopy Millimeter and submillimeter absorption spectra	Lasar raman spectroscopy Millimeter and submillimeter wave absorption spectra. Direct absorption of IR quanta in amide-I bond

Acknowledgments

These studies were supported by Department of Energy Contract DE-AIO1-79 ET 29078, Bureau of Radiological Health Contract FDA RO1-FD 00963-03, and by the Southern California Edison Company. We are grateful to Ms. Karen Kapciuk for preparation of illustrations and Ms. Janice Merrill for her care in preparation of the manuscript.

References

Adey, W. R. (1977) Models of membranes of cerebral cells as substrates for information storage. *Biosystems* 8: 163–178.

Adey, W. R. (1981a) Tissue interactions with nonionizing electromagnetic fields. *Physiol. Rev.* 61: 435–514.

Adey, W. R. (1981b) Ionic nonequilibrium phenomena in tissue interactions with electromagnetic fields. In: *Biological Effects of Nonionizing Radiation*, K. H. Illinger, ed. Washington, D.C.: American Chemical Soc. Symposium Series, No. 157, pp. 271–297.

Bawin, S. M., and Adey, W. R. (1976) Sensitivity of calcium binding in cerebral tissue to weak environmental electric fields oscillating at low frequency. *Pro. Natl. Acad. Sci. USA.* 73: 1999–2003.

Bawin, S. M., Adey, W. R., Lawrence, A. F., Bassen, H. I. (1980) Weak amplitude-modulated radiofrequency fields modify $^{45}Ca^{2+}$ release from cat cerebral cortix *in vivo.* Bioelectromagnetics Society, 2nd Annual Meeting. *Bioelectromagnetics* 1: 211 (abstract No. 78).

Bawin, S. M., Kaczmarek, L. K., and Adey, W. R. (1975) Effects of modulated VHF fields on the central nervous system. *Ann. NY Acad. Sci.*, 247: 74–81.

Bawin, S. M., Sheppard, A. R., and Adey, W. R. (1978) Possible mechanism of weak electromagnetic field coupling in brain tissue. *Bioelectrochem. Bioenergetics* 5: 67–76.

Bishop, A. R., Krumhausl, J. A., and Trullinger, S. E. (1980) Solitons in condensed matter: a paradigm. *Phusical S.* 1: 1–44.

Blackman, C. S., Elder, J. A., Weil, C. M., Beunane, S. G., Eichinger, D. C., and House, D. E. (1979) Induction of calcium ion efflux from brain tissue by radio frequency radiation: effects of modulation frequency and field strength. *Radio Sci.* 14: 93–98.

Borle, A. E. (1969) Kinetic analysis of calcium movements in HeLa cell cultures. *J. Gen. Physiol.* 53: 43–56.

Borle, A. E., and Anderson, J. H. (1976) A cybernetic view of cell calcium metabolism. In: *Symposia of the Society for Experimental Biology, No. 30, Calcium in Biological Systems*, O. J. Duncan, ed. London: Cambridge University Press, pp. 141–160.

Bullough, R. K. (1977) International Atomic Energy Agency SMR 20/51, pp. 381–469.

Cotterill, R. M. J. (1980) Phase transition of the lipid bilayer. *Physica Scripta* 22: 188–192.

Davidson, R. C. (1972) *Methods in Nonlinear Plasma Theory.* New York: Academic Press.

Davydov, A. S. (1971) *Theory of Molecular Excitons*, New York: Plenum Press.

Davydov, A.S. (1973) The theory of contractions of proteins under their excitation. *J. Theor. Biol.* 38: 559–569.

Davydov, A. S. (1977) Solitons as energy carriers in biological systems. *Studia Biophysica* 62: 1–8.

Davydov, A. S. (1979a) Solitons in molecular systems. *Physica Scripta* 20: 387–394.

Davydov, A. S. (1979b) *Biology and Quantum Mechanics.* Kiev:Nauka.

Davydov, A. S. (1979c) Nonlinear vibrational phenomena in biology. *Acad. Sci. Ukrainian SSR, Inst. for Theor. Phys.*, Preprint ITP-79-69E.

Davydov, A. S. (1980) Influence of electro-phonon interaction on the motion of an electron in one dimensional molecular system. *Theor. Math. Physics* 40: 831–840.

Davydov, A. S., and Kislukha, N. I. (1976) Solitons in one-dimensional molecular chains. *Zh. Eksp. Teor. Fiz.*, 71: 1090–1098.

Dmitriev, V. G., Emelovanov, V. N., Kashintev, M. A., Kulikov, V. V., Solikoev, A. A., Stelomakh, M. F., and Cherednichenko, O. B. (1979) Nonlinear perception of infrared radiation in the 800–1355 nm range with human eye. (Tr.) *Sov. J. Quantum Electronics* 9: 475–479.

Fife, P.C. (1977a) Stationary patterns for reaction-diffusion equations. In *Nonlinear diffusion*, W.E. Fitzgibbon and H.F. Walker, eds. San Francisco: Pitman, 81–121.

Fife, P. C. (1977b) Asymptotic analysis of reaction-diffusion wave fronts. *Rocky Mountain J. Math.* 7: 389–415.

Fullmer, C. S., and Wasserman, R. H. (1979) Analytical peptide mapping by high performance liquid chromatography: application to intestinal calcium-binding proteins. *J. Biol. Chem.* 254: 7208–12.

Gardiner-Medwin, A. R. (1980) Membrane transport to solute migration affecting the brain cell microenvironment. *Neurosci. Res. Prog. Bull.* 18: 208–226.

Glansdorff, P., and Prigogine, I. (1971) *Thermodynamic Theory of Structure, Stability and Fluctuations.* New York: Wiley-Interscience.

Green, D. E. (1974) A framework of principles for the unification of bioenergetics. *Ann. New York Acad. Sci.* 227: 6–45.

Hodgkin, A. L., and Huxley, A. F. (1952) A quantitative description of membrane current and its application to conduction and excitation in nerve. *J.Physiol.* (London) 234: 500–544.

Hyman, J. M., McLaughlin, D. W., and Scott, A. C. (1981) On Davydov's alpha-helix soliton. *Physica* 30: 23–44.

Jacobson, K., and Papahadjopoulos, D. (1975) Phase transitions and phase separations in phospholipid membranes induced by changes in temperature, pH, and concentration of bivalent cations. *Biochem.* 14: 152–161.

Kendig, J. J. (1981) Anesthetics and pressure in nerve cells. In: *Molecular Mechanisms of Anesthesia*, Vol. 2, B. R. Fink, ed. New York: Raven Press.

Keynes, R. D. (1970) Evidence for structural changes during nerve activity and their relation to the conduction mechanism. In: *The Neurosciences; Second Study Program.* F. O. Schmitt, ed. New York: Rockefeller University Press, pp. 707–714.

Kopell, N. (1977) Waves, shocks and target patterns in an oscillating chemical

reagent. In: *Nonlinear Diffusion*, W. E. Fitzgibbon and H. F. Walker, eds. San Francisco: Pitman, pp. 129–154.

Kopell, N. (1978) Reaction-diffusion equations and pattern formation. In: *Studies in Mathematical Biology*, S. A. Levin, ed. Washington, D. C.: Math. Assn. of America, pp. 191–205.

Lazarides, E. (1980) Intermediate filaments as mechanical integrators of cellular space. *Nature* 283: 249–256.

Levine, B. A., Williams, R. J. P., Fullmer, C. S., and Wasserman, R. H. (1979) NMR studies of various calcium-binding proteins. In: *Calcium Binding Proteins and Calcium Function* R. H. Wasserman *et al.*, eds. New York: Elsevier, pp. 29–37.

Lin-Liu, S., and Adey, W. R. (1980) Effect of ELF modulated 450 MHz field on calcium efflux from rat synaptosomes. Proc. Second Annual Meeting, Bioelectromagnetics Society. *Bioelectromagnetics* 1: 211 (abstract).

Low, P. S., Lloyd, D. H., Stein, T. M., and Rogers, J. A. (1979) Calcium displacement by local anesthetics. *J. Biol. Chem.* 254: 4119–4125.

Marcuse, D. (1980) *Principles of Quantum Electronics* New York: Academic Press.

Matsumoto, G., and Sakai, H. (1979) Microtubules inside the plasma membrane of squid giant axons and their possible physiological functions. *J. Mem. Biol.* 50: 15–22.

McClain, D. S., and Edelman, G. M. (1978) Surface modulation and transmembrane control. In: *The Molecular Basis of Cell-Cell Interaction*, R. A. Lerner and D. Bergsma, eds. New York: Alan R. Liss, Inc. pp. 1–28.

Melchior, D. L., and Stein, J. M. (1979) Lipid-associated thermal events in biomembranes. *Progress in Surface and Membrane Science*, D. A. Cadenhead and J. F. Danielli, eds. New York: Academic Press. 13: 211–296.

Moore, W. J. (1972) *Physical Chemistry.* Englewood Cliffs, N.J.: Prentice-Hall, pp. 449–454.

Nazarea, A. D. (1979) Electric fields and self-coherent patterns and structures in chemical systems: large-scale effects and biological implications. *J. Chem. Physics* 9: 415–449.

Nicholson, C. (1977) Brain-cell microenvironment as a communication channel. In: *The Neurosciences; Fourth Study Program.* F. O. Schmitt and F. C. Worden, eds. Cambridge: MIT Press, pp. 457–476.

Nicholson, C. (1980) Dynamics of the brain cell microenvironment. *Neurosci. Res. Program Bull.* 18: 275–289.

Nicholson, C., ten Bruggencate, G., Steinberg, R., and Stoeckle, H. (1977) Calcium modulation in brain cell extracellular environment demonstrated with ion-selective micropipette. *Pro. Natl. Acad. Sci.* USA. 74: 1287–1290.

Nicolis, G., and Prigogine, I. (1977) *Self-Organization in Nonequilibrium Systems.* New York: Wiley.

Schmidt, S., and Ortoleva, P. (1977) A new chemical wave equation for ionic systems. *J. Chem. Physics* 67: 3771–3776.

Schmidt, S., and Ortoleva, P. (1978) Multiple chemical waves induced by applied electric field. *J. Chem. Physics* 67: 1010–1015.

Scott, A. C. (1981) The laser-Raman spectrum of a Davydov soliton. *Physics Letters* 86A: 60–62.

Scott, A. C., Chu, F. Y. F., and McLaughlin, D. W. (1974). The soliton: a new concept in applied science. Proc. IEEE 61: 1443–1483.

Seamon, K. B. (1980) Calcium and magnesium-dependent conformational states of calmodulin as determined by nuclear magnetic resonance. *Biochem.* 19: 207–215.

Tasaki, I. (1974) Energy transduction in the nerve membrane and studies of excitation processes with extrinsic fluorescence probes. *Ann. N.Y. Acad. Sci.* 227: 147–267.

Tasaki, I. (1975) Evolution of theories of nerve excitation. In: *The Nervous System, Volume 1: The Basic Neurosciences* D. B. Tower, ed. New York: Raven Press, pp. 177–195.

Taylor, L. S. (1981) Athermal millimeter/microwave biological effects. *Bioelectromagnetics.* 2: 259–268.

Triffet, T., and Green, H. S. (1980) Information and energy flow in a simple nervous system. *J. Theor. Biol.* 86: 3–44.

Troy, W. C., and Field, R. J. (1977) The amplification before decay of perturbations around stable states in a model of the Zhabotinskii reaction. SIAM *J. App. Math.* 32: 306–322.

Trullinger, S. E. (1978) Where do solitons go from here? In: *Solitons and Condensed Matter Physics* A. R. Bishop and T. Schneider, eds. Berlin: Springer-Verlag, pp. 338–340.

Urry, D. W. (1978) Basic aspects of calcium chemistry and membrane interaction: on the messenger role of calcium. *Ann New York Acad. Sci.* 307: 3–27.

Vaccaro, S. R., and Green, H. S. (1979) Ionic processes in excitable membranes. *J. Theor. Biol.* 81: 771–802.

Webb, S. J. (1980) Laser-Raman spectroscopy of living cells. *Physics Reports* 60: 201–224.